

Course: SER216

Student Name: Elyor Ataboyev

Project: Digital Parking Permit System

GitHub Repository: https://github.com/Elyor200/SER216_Final_Project_ElyorAtaboyev.git

Task 1: Requirements

Functional Requirements:

1. The system shall allow students and staff to register vehicle details (make, model, license plate number) linked to their university ID.
2. The system shall enable users to submit a parking permit request for a specific duration and permit type.
3. The system shall display real-time availability of parking zones based on the user's permit eligibility.
4. The system shall allow security staff to scan/verify permits via mobile device and log parking violations for unauthorized vehicles.

Non-Functional Requirements:

1. **Performance:** The permit verification process at security checkpoints must complete within 2 seconds under normal system load.
2. **Security:** All personal user data and vehicle registration information must be encrypted at rest and in transit using AES-256 encryption standards.

Task 2: Workflow Diagram



Workflow Description:

- 1. **Vehicle Registration:** User inputs vehicle details into the system.
- 2. **Permit Request:** User submits a request for a parking permit.
- 3. **Zone Check:** System verifies eligibility and displays available parking zones.
- 4. **Permit Approval:** System or admin approves the request and issues the digital permit.
- 5. **Parking Verification:** Security staff scans the permit to validate access or log violations.

Task 3: Test Case Table

Test ID	Feature	Input / Data	Expected Result	Test Type
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TC-01	Integrated Permit & Zone Check	Valid user requests permit for Zone A	System assigns permit and decreases Zone A capacity by 1	System Test
TC-02	Security Verification	Security scans an invalid/unregistered license plate	System flags violation and updates violation record	System Test
TC-03	User Registration	Staff member registers vehicle with valid University ID	Account created and staff-specific permit options displayed	Acceptance Test
TC-04	Expiration Notification	Permit expiration date is within 3 days	User receives automated email notification to renew permit	Acceptance Test
TC-05	Vehicle Registration Module	Submit new vehicle details after recent DB schema update	Vehicle saved correctly without data loss from previous modules	Regression Test
TC-06	Permit Verification Logic	Scan a known valid, unexpired permit	System displays "Valid" and does NOT flag as expired	Regression Test

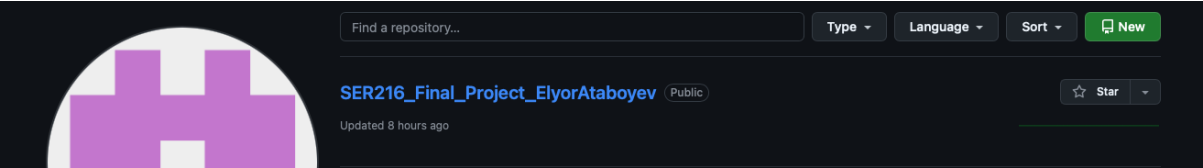
Task 4: Defect Analysis

Defect: "A valid parking permit is shown as expired during security verification."

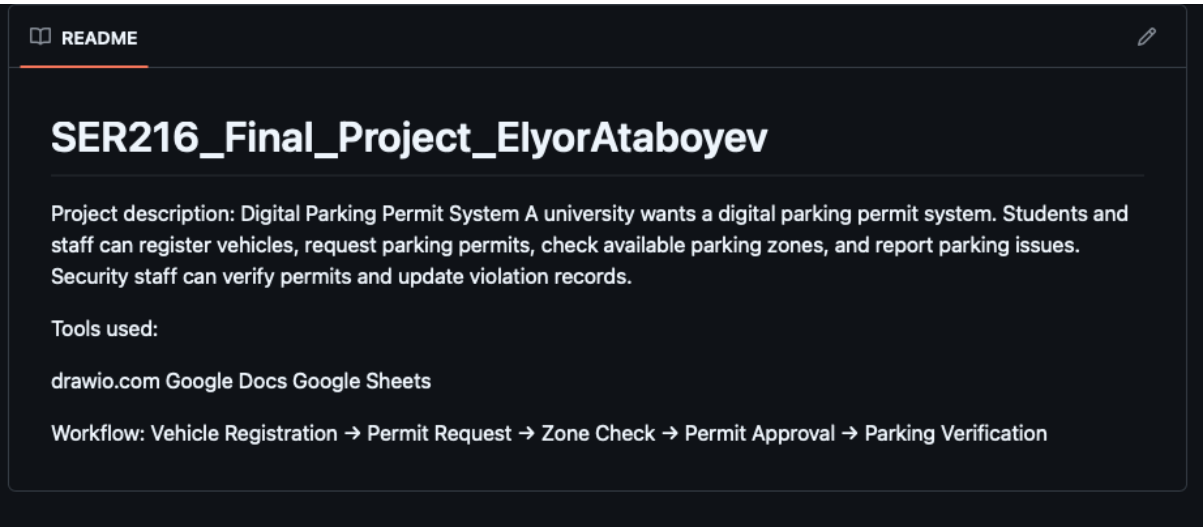
Item	Student Answer
Defect category	Functional / Logic Error
Severity	High (Prevents valid users from parking and generates false violations)
Priority	High (Must be fixed immediately before system launch/go-live)
Possible root cause	Timezone mismatch between the server database and the security scanner device, OR incorrect date comparison logic in the verification API (e.g., using <code>expiration_date < current_date</code> instead of <code><=</code> , or comparing dates without accounting for time-of-day).
Regression test	TC-06: After applying the fix, scan a known valid, unexpired permit to confirm the system correctly displays "Valid" and does not trigger a false expiration flag.

Task 5: GitHub Evidence

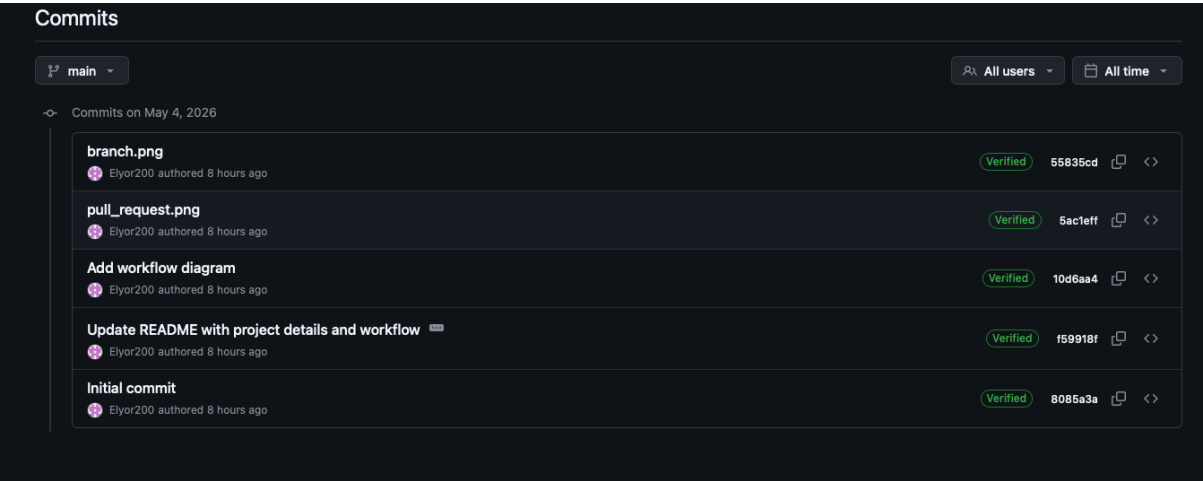
1. Repository Homepage



2. README.md File



3. Commit History (Showing at least 3 commits)



4. Feature Branch

Branches

New branch

OverviewYoursActiveStaleAll

Q Search branches...

Default

Branch	Updated	Check status	Behind	Ahead	Pull request
main	now				Default

Your branches

Branch	Updated	Check status	Behind	Ahead	Pull request
feature/change-request	3 minutes ago		1	1	
feature/test-plan	8 minutes ago		2	0	

Active branches

Branch	Updated	Check status	Behind	Ahead	Pull request
feature/change-request	3 minutes ago		1	1	
feature/test-plan	8 minutes ago		2	0	

5. Pull Request

Add workflow diagram #1

Merged Elyor200 merged 1 commit into feature/change-request from main 1 minute ago

Conversation 0Commits 1Checks 0Files changed 1

Elyor200 commented 1 minute ago

No description provided.

Add workflow diagram

Elyor200 merged commit 7f357c8 into feature/change-request 1 minute ago

Pull request successfully merged and closed

You're all set — the branch has been merged.

Add a comment

WritePreview

Add your comment here...

Markdown is supportedPaste, drop, or click to add files

Comment

6. Merged Pull Request

The screenshot shows a GitHub Pull Request interface. At the top, the title is "Add workflow diagram #1" with an edit icon. Below the title, a status bar indicates "Merged" in a purple box, followed by "Elyor200 merged 1 commit into feature/change-request from main" and a clock icon with "1 minute ago".

Below the status bar, there are tabs for "Conversation" (0), "Commits" (1), "Checks" (0), and "Files changed" (1). The "Conversation" tab is active, showing a comment from Elyor200: "No description provided." with a "1 minute ago" timestamp and an "Owner" label.

Below the comment, a commit history section shows a commit "Add workflow diagram" by Elyor200, marked as "Verified" with a green checkmark and the hash "10d6aa4". Below this, a message states "Elyor200 merged commit 7f357c8 into feature/change-request" with a "1 minute ago" timestamp and a "Revert" button.

A large green box in the center contains the message: "Pull request successfully merged and closed. You're all set — the branch has been merged."

At the bottom, there is a section for "Add a comment" with a text area, a "Write" tab, a "Preview" tab, and a rich text editor toolbar. The text area contains the placeholder "Add your comment here...". Below the text area, there are links for "Markdown is supported" and "Paste, drop, or click to add files". A green "Comment" button is at the bottom right.

On the right side, there is a sidebar with various sections: "Reviews" (No reviews), "Assigned" (No one assigned), "Labels" (None yet), "Projects" (None yet), "Milestones" (No milestones), "Development" (Successful issues), "Notifications" (You're required to open/close), and "Participants" (1 participant).

7. Uploaded Final PDF in Repository